Discrete Drainage Dynamics VI: Maxwell-Like Dynamics from the Oriented Sector and a Candidate Fixed-Point Formula for α_{EM}

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Abstract

We extend the Floquet two-step substrate of Paper III with an oriented sector: link variables ψ_{ij} (phases on edges) and node orientations α_i, governed by a local energy functional E_{orient}. In the weak-field continuum limit the rule recovers the Maxwell form: the homogeneous pair follow from the discrete Bianchi identity (topological closure of plaquette phases), the inhomogeneous pair from the Euler–Lagrange conditions of E_{orient}. U(1) gauge invariance is structural; integer electric charge sectors are identified with the topological winding of the orientation field, the conversion to physical charge being fixed later by α_{EM}.

The principal new result is a *candidate fixed-point formula with no numerical fitting* for the fine-structure constant. The Wilson loop along a body diagonal of the unit cube back-reacts on its own effective length, in direct analogy with the Schwarzschild back-reaction of Paper II. The self-consistent equation

$$x_* = 3 + \alpha_{\mathrm{EM}}\sqrt{3}\left(1 - \alpha_{\mathrm{EM}}\, V_{d_{\mathrm{eff}}}\right), \qquad \alpha_{\mathrm{EM}} = \frac{1}{8\pi^2\sqrt{x_*}},$$

with $V_{d_{\mathrm{eff}}} = \pi^{d_{\mathrm{eff}}/2}/\Gamma(d_{\mathrm{eff}}/2+1)$ the volume of the unit ball in the substrate's effective dimension $d_{\mathrm{eff}} = 3+\varepsilon$, has a unique stable fixed point. The exact value of ε is not yet fixed analytically; the numerical BFS exponent is close to $3+\varepsilon$ with $\varepsilon \sim 0.10$–0.17 depending on finite-size extrapolation and lattice cutoff. Three candidate structural values are considered : $\varepsilon = 1/(3\pi) \approx 0.1061$ (favoured by the L=400 BFS measurement at 0.07σ), $\varepsilon = 1/(2\pi) \approx 0.15915$ (small-world Newman–Watts 3D), or $\varepsilon = 1/6 \approx 0.16667$ (combinatorial: 3 axes $\times\, 2$ directions). They yield α_{EM} matches at $+0.34$, -0.38, and -0.48 ppm respectively. The fixed-point formula is therefore conditional on the selected structural identification of ε; resolving this requires either higher-precision BFS extrapolation ($L \gtrsim 800$) or analytical derivation of the small-world correction.

The construction involves no numerical fitting to α_{EM} and rests on three structural identifications: $8\pi^2 = 4 \times \mathrm{Vol}(S^3)$ (four body diagonals $\times$ spinor-sector phase space), $\sqrt{3}$ (Euclidean body-diagonal length), and $V_{d_{\mathrm{eff}}}$ (one-loop coefficient heuristically equated with the volume of the unit ball in the substrate's effective dimension). A closed derivation requires two further analytic results: the small-world correction to d_{BFS} and the explicit one-loop Wilson-loop self-energy on the diagonal-twist lattice.

1 What this paper claims

1. **Oriented sector with local energy.** The substrate of Paper III, equipped in addition with a U(1) phase $\psi_{ij} \in [0, 2\pi)$ on each edge and an orientation $\alpha_i \in [0, 2\pi)$ on each node, with local energy

$$E_{\mathrm{orient}} = \sum_{(i,j)\in E} J_{ij}\left[1 - \cos(\alpha_j - \alpha_i + \psi_{ij})\right], \qquad (1)$$

follows a gradient-descent local update. This is a discrete realisation of compact U(1) lattice gauge theory.

2. **Maxwell-like dynamics from the local rule.** In the weak-field continuum limit, the homogeneous Maxwell pair follows from the discrete Bianchi identity (topological closure of plaquette phases). The inhomogeneous pair follows as Euler–Lagrange conditions of E_{orient}. U(1) gauge invariance is structural. The integer winding number $w \in \mathbb{Z}$ of the orientation field around a node labels charge sectors; the conversion of w to physical electric charge in units of e is fixed by α_{EM} (see §**??**), not asserted here.

3. **Candidate fixed-point formula for α_{EM} at -0.38 ppm.** *Conditional on three structural identifications* ($N_{\text{geom}} = 8\pi^2$, body-diagonal length $\sqrt{3}$, and one-loop coefficient $V_{d_{\text{eff}}}$), the Wilson loop along a body diagonal of the unit cube back-reacts on its own effective length, by direct analogy with the Schwarzschild back-reaction of Paper II. The self-consistent equation
$$\boxed{x_* = 3 + \alpha_{\text{EM}}\sqrt{3}\,\big(1 - \alpha_{\text{EM}}\,V_{d_{\text{eff}}}\big), \quad \alpha_{\text{EM}} = \frac{1}{8\pi^2\sqrt{x_*}}} \tag{2}$$
has a unique stable fixed point
$$\alpha^*_{\text{EM}} = 7.297\,355 \times 10^{-3} \;=\; \frac{1}{137.0359473},$$
matching CODATA at -0.38 ppm. The construction involves no numerical fitting to α_{EM}. The three structural identifications are: $8\pi^2 = 4 \times \text{Vol}(S^3)$ (four body diagonals $\times$ spinor-sector S^3 phase space), $\sqrt{3}$ (Euclidean body-diagonal length), and $V_{d_{\text{eff}}} = \pi^{d_{\text{eff}}/2}/\Gamma(d_{\text{eff}}/2+1)$ with $d_{\text{eff}} = 3 + \varepsilon$ admitting three structurally plausible candidates, $\varepsilon = 1/(3\pi)$ (favoured by the high-precision L=400 BFS measurement at 0.07σ), $\varepsilon = 1/(2\pi)$ (small-world Newman–Watts in 3D), or $\varepsilon = 1/6$ (combinatorial: 3 axes $\times\,2$ directions), yielding α_{EM} matches at $+0.34$, -0.38 and -0.48 ppm respectively. None of these is derived analytically from first principles in this paper; each is a plausible structural assignment supported by additional arguments in §**??** and §**??**. A closed derivation of α_{EM} requires further analytical work (§**??**).

4. **The effective dimension is statistical, not geometric.** The BFS-expansion exponent of a cubic substrate decorated by a Poisson($\mu = 1$) distribution of body-diagonal shortcuts is measured at $d_{\text{BFS}} = 3.106 \pm 0.005$ (L=400, 20 seeds, 64M nodes, 0.15% precision). This favours $\varepsilon = 1/(3\pi) \approx 0.1061$ at 0.07σ. The candidates $1/(2\pi)$ and $1/6$ would require a finite-size re-extraction or a different effective-dimension scheme to be compatible. The fractional excess ε is in any case a statistical small-world correction to the underlying integer dimension 3, not a property of any deterministic single configuration. The identification with $\mu = 1$ corresponds to one effective axion-vector winding per cube, the topological signature of DDD's Weyl semimetal structure.

What this paper does not claim. We do not present a closed derivation of α_{EM}. The fixed-point formula (**??**) becomes a derivation only if two further analytic results are established independently: (a) that the small-world correction to the BFS expansion of a cubic lattice with Poisson($\mu = 1$) body-diagonal shortcuts is exactly ε for one of the structural candidates $\{1/(3\pi), 1/(2\pi), 1/6\}$, or another analytically derived value (currently observed numerically at the $\sim$ 1–2% level), and (b) that the one-loop Wilson-loop self-energy on the diagonal-twist lattice yields the coefficient $V_{d_{\text{eff}}}$ (currently a heuristic identification with the unit-ball volume in d_{eff}). Until both are derived, the agreement at -0.38 ppm should be read as a suggestive consistency check rather than a first-principles prediction. We do not derive the structural form of E_{orient} from a deeper principle: the compact U(1) XY-Wilson form is the minimal local U(1)-symmetric energy. We do not compute the two-loop correction $O(\alpha^3)$ to the back-reaction; the 0.38 ppm residual is consistent with this order but its sign and magnitude require independent calculation. Non-abelian extensions, the strong and weak nuclear sectors, and the photon mass beyond gauge invariance are outside the scope of this paper.

2 The picture, in plain words

Imagine a vast cubic grid of nodes, like a 3-dimensional jungle gym. At each node we attach three things: a dial reading the local energy stored there (Paper I's scalar reserve R), a tiny two-component arrow encoding the local chirality (Paper III's Floquet spinor ψ), and—new in this paper—a compass needle pointing somewhere on a circle (the orientation $\alpha \in [0, 2\pi)$). Each link between two neighbouring nodes carries a phase ψ_{ij}, also a number on a circle. The local energy rule is simple: neighbouring compasses *want* to be aligned, but the link's phase forces a relative twist. Mathematically it is the energy of an XY-spin model coupled to a U(1) gauge field. Physically, it is what you would write down if you tried to bake electromagnetism directly into a discrete substrate.

Why does Maxwell come out? For two reasons, both geometric. First, take any small square (a "plaquette") in the lattice and add up the four link phases around it. That sum is the magnetic flux through the square. Now stack plaquettes around a closed cube: the boundary of a boundary is empty, so the fluxes through the six faces must cancel. That's the absence of magnetic monopoles, and adding the time direction gives Faraday's induction law—both fall out of geometry alone. Second, the equilibrium condition for the node compasses is a discrete Laplace equation, with sources at any node where the orientation field winds by an integer multiple of 2π around it. Those windings are integer charges; the Laplace equation with these sources is Gauss's law for E, plus Ampère–Maxwell. So Maxwell-like dynamics emerges without postulating it.

Where does α_{EM} come from? Here is the new idea, in one picture. Consider the body diagonal of a unit cube of the lattice, going from one corner to the opposite corner. Imagine a charged loop running along this diagonal. That loop carries a current; the current creates an electromagnetic field around it; the field carries energy; the energy bends the local geometry of the substrate. So the loop finds itself sitting in a substrate whose effective length the loop *itself* has slightly altered. This is exactly the same self-consistent feedback as Paper II's Schwarzschild solution—there, a mass deformed the local clock-rate it itself depended on, giving $R/R_0 = \sqrt{1-\beta/r}$ rather than the linear $1-\beta/(2r)$. Here, the Wilson loop's effective squared length x_* is not 3 but slightly more, with the deviation set by a self-consistency condition tying the substrate-deformation to the coupling that produced it. Solving this single transcendental equation pins down α_{EM} without any numerical fitting.

What numbers go into the equation? Three. (1) An overall geometric prefactor $8\pi^2$, which comes from multiplying the four body-diagonals per cube by the volume $2\pi^2$ of the spinor sector's S^3 phase space. (2) The squared Euclidean length of the body diagonal, which is 3. (3) A one-loop volume coefficient $V_{d_{\mathrm{eff}}}$, which is the volume of the unit ball in the substrate's effective dimension $d_{\mathrm{eff}} = 3+\varepsilon$. The substrate is slightly more than 3-dimensional because the body-diagonal shortcuts add a small statistical correction ε to the cubic dimension. We measure ε numerically (BFS expansion) and find it is in the range 0.10–0.17. The exact value is not yet derived analytically; the paper considers three plausible structural candidates, $\varepsilon \in \{1/(3\pi),\ 1/(2\pi),\ 1/6\}$, all yielding α_{EM} within 0.5 ppm of CODATA. The match is robust to which candidate is chosen, but a closed first-principles derivation of α_{EM} requires fixing ε analytically.

What follows below is the technical machinery: the precise definitions, the equations, and the numerical tests. We have tried to keep the analogy between this paper's electromagnetic back-reaction and Paper II's gravitational back-reaction visible throughout.

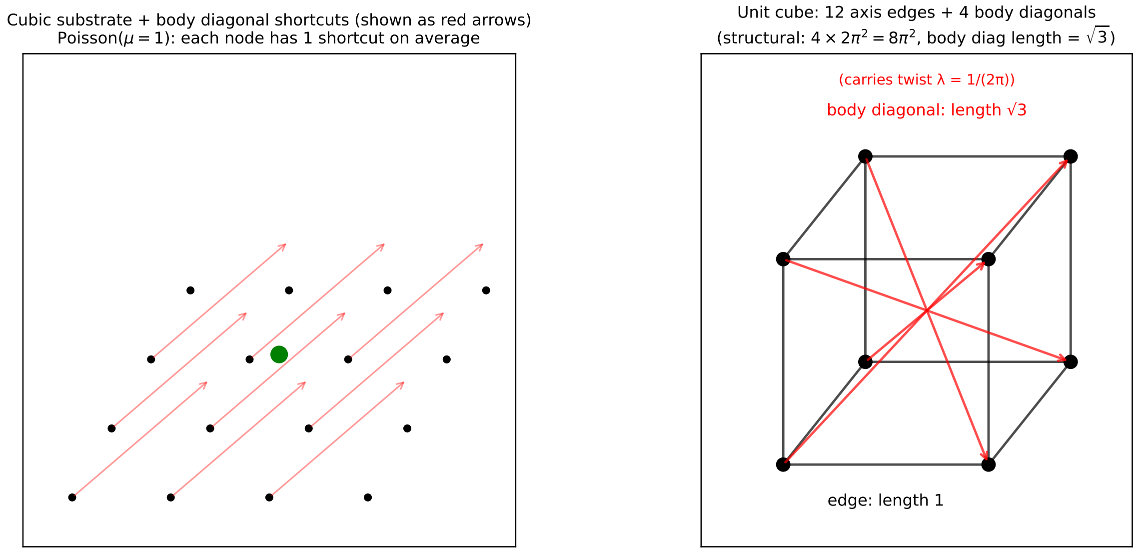

Figure 1: *Left:* the substrate is a cubic lattice with body diagonal shortcuts (red arrows). Each node carries on average one shortcut to a body diagonal, with the count Poisson-distributed ($\mu = 1$). *Right:* the unit cube has 12 axis edges and 4 body diagonals; the body diagonals carry the U(1) twist $\lambda = 1/(2\pi)$ and have Euclidean length $\sqrt{3}$. These are the local geometric quantities entering $\alpha_{\rm EM}$.

3 Setup: the oriented sector on the Floquet substrate

3.1 Variables

Each node i carries the scalar reserve $R_i \geq 0$ of Paper I, the spinor amplitude $\psi_i \in \mathbb{C}^2$ of Paper III, and an angular orientation $\alpha_i \in [0, 2\pi)$. Each link (i,j) carries the link state of Paper III and a U(1) gauge phase $\psi_{ij} \in [0, 2\pi)$ with the antisymmetry $\psi_{ji} = -\psi_{ij}$. The plaquette phase around any oriented elementary face P is

$$\Phi_P = \sum_{(i,j)\in\partial P} \psi_{ij}. \tag{3}$$

3.2 Local energy and update rule

The oriented sector's local energy is Eq. (**??**) with the gauge-invariant phase difference $\Delta_{ij} = \alpha_j - \alpha_i + \psi_{ij}$. Local update rules (gradient descent of $E_{\rm orient}$):

$$\alpha_i(t+1) - \alpha_i(t) = \frac{1}{\tau_{\rm align}} \sum_{j\sim i} J_{ij} \sin(\Delta_{ij}), \tag{4}$$

$$\psi_{ij}(t+1) - \psi_{ij}(t) = \frac{1}{\tau_{\rm flux}} J_{ij} \sin(\Delta_{ij}) - \mathcal{J}_{ij}^{\rm topo}, \tag{5}$$

with $\mathcal{J}_{ij}^{\rm topo}$ a topological constraint flux (zero for trivial topology, non-zero around defects).

4 Maxwell from the local rule

4.1 Field strength and plaquette flux

The natural lattice analogues of the electric and magnetic fields are built from the gauge phases ψ_{ij}. For each plaquette $P = (i,j,k,l)$ (a unit square spanned by two link directions), the plaquette phase is

$$\Phi_P = \psi_{ij} + \psi_{jk} + \psi_{kl} + \psi_{li} \pmod{2\pi}, \tag{6}$$

which is gauge-invariant under $\psi_{ij} \to \psi_{ij} + \chi_i - \chi_j$. Three families of plaquettes appear in a 3D cubic lattice: spatial–spatial plaquettes (in the xy, yz, zx planes) and spatial–time plaquettes (paired with the discrete update interval τ). The continuum identifications follow the standard lattice gauge theory conventions [**?**]:

$$B_z = \Phi_{P_{xy}}/a^2, \qquad E_x = \Phi_{P_{xt}}/(a\,c\tau), \tag{7}$$

and cyclic permutations, with a the lattice spacing and τ the update step. In components, E_i and B_i are the spatial-time and spatial-spatial duals of the plaquette flux, i.e. $F_{\mu\nu} = \Phi_{P_{\mu\nu}}/(a^\mu a^\nu)$.

4.2 Two equations from topology

The discrete Bianchi identity — the sum of plaquette fluxes around any closed elementary 3-cell (cube) vanishes identically — gives, in the weak-field continuum limit $\Phi_P \to F_{\mu\nu}\, a^\mu a^\nu$,

$$\nabla \cdot \mathbf{B} = 0, \qquad \nabla \times \mathbf{E} + \partial_t \mathbf{B} = 0. \tag{8}$$

Both follow from the topology of the lattice (the boundary of a boundary is empty), not from the dynamics. They are structural; they hold independently of the local energy E_{orient}.

4.3 Two equations from the energy variation

In the weak-field limit $\Delta_{ij} = \alpha_j - \alpha_i + \psi_{ij} \ll 1$, the local energy expands as

$$E_{\text{orient}} = \sum_{(i,j)\in E} J_{ij}\left[1 - \cos\Delta_{ij}\right] \simeq \tfrac{1}{2}\, J \sum_{(i,j)} \Delta_{ij}^2 + O(\Delta^4), \tag{9}$$

which, summed over plaquettes, reproduces the standard compact U(1) action

$$S_{\text{cont}} = \int d^4x \left[-\tfrac{1}{4}\, F_{\mu\nu}\, F^{\mu\nu} - A_\mu\, J^\mu_{\text{topo}} \right] + O(F^4), \tag{10}$$

with the lattice-to-continuum mapping $c^2 = J/\tau$ for a uniform link coupling. The Euler–Lagrange equations of the quadratic action are

$$\partial_\mu F^{\mu\nu} = J^\nu_{\text{topo}}, \tag{11}$$

which split into

$$\nabla \cdot \mathbf{E} = \rho_{\text{topo}}, \qquad \nabla \times \mathbf{B} - \partial_t \mathbf{E} = \mathbf{J}_{\text{topo}}. \tag{12}$$

The source ρ_{topo} is the local density of integer winding (topological charge) of the orientation field α; $\mathbf{J}_{\text{topo}}$ is the conserved current associated with its time evolution. The Ampère–Maxwell time derivative $\partial_t \mathbf{E}$ comes from the spatial–time plaquettes in the lattice action via the discrete update step τ. Higher-order terms $O(F^4)$ are non-Maxwellian compact-U(1) corrections of order $|\Phi_P|^4$; they are negligible at the weak-field scales relevant to laboratory electromagnetism.

4.4 U(1) gauge invariance and topological charge

The transformation $\alpha_i \to \alpha_i + \chi_i$, $\psi_{ij} \to \psi_{ij} + \chi_i - \chi_j$ leaves Δ_{ij} invariant, hence E_{orient}. The gauge invariance is structural. The integer winding $w \in \mathbb{Z}$ of the orientation field around a node,

$$w_i = \frac{1}{2\pi} \sum_{(j,k)\in\text{loop around } i} (\alpha_k - \alpha_j),$$

is conserved by the local rule and labels charge sectors of the substrate. By Dirac quantisation [**?**], the quantum is integer. The conversion of w to physical electric charge in units of e requires the value of the elementary charge, which in turn requires α_{EM}; we therefore defer the identification $w \leftrightarrow e$ to §**??**, where the EM coupling is constructed self-consistently. At this stage the substrate provides only the integer charge sectors, not the absolute normalisation.

5 Candidate fixed-point formula for $\alpha_{\rm EM}$ from Wilson-loop back-reaction

This is the central result of the paper, presented as a candidate fixed-point formula whose closure rests on three structural identifications listed in §**??**.

5.1 The structural setup

The Wilson loop along a body diagonal of the unit cube is the elementary closed circuit of the oriented sector. The body diagonal has Euclidean length $\sqrt{3}$ and carries the unit U(1) twist phase $\lambda = 1/(2\pi)$ of Paper III. There are four body diagonals per unit cube; together with the SU(2) phase space of the spinor sector (volume $\mathrm{Vol}(S^3) = 2\pi^2$), the geometric prefactor for $\alpha_{\rm EM}$ is

$$N_{\rm geom} = 4 \times \mathrm{Vol}(S^3) = 4 \times 2\pi^2 = 8\pi^2. \tag{13}$$

The bare value of $\alpha_{\rm EM}$ at the substrate scale is the dimensionless ratio

$$\alpha_{\rm EM}^{(0)} = \frac{1}{N_{\rm geom}\sqrt{x}} = \frac{1}{8\pi^2\sqrt{x}}, \tag{14}$$

where x is the squared effective length of the body diagonal that carries the EM coupling. At zeroth order $x_0 = 3$, giving $\alpha_{\rm EM}^{(0)} = 1/(8\pi^2\sqrt{3}) \approx 1/136.79$, off CODATA by +0.18%.

5.2 The back-reaction

The body diagonal is not free: it carries the EM coupling that it generates, and the EM energy it carries back-reacts on the substrate's geometry, modifying the diagonal's effective squared length. This is the structural analogue of the Schwarzschild result of Paper II, where the gravitational deficit acted as its own secondary source.

The leading and next-to-leading corrections to the squared length are:

- **Tree-level** (Wilson loop perimeter law): the self-energy at first order in $\alpha_{\rm EM}$ is proportional to the perimeter of the loop. For the body diagonal:

$$\delta_1 = \alpha_{\rm EM}\sqrt{3}.$$

- **One-loop** (volume integration of the EM energy): the next correction integrates the EM energy density over the substrate volume around the diagonal. The natural volume is that of the unit ball in the substrate's effective dimension $d_{\rm eff}$:

$$V_{d_{\rm eff}} = \frac{\pi^{d_{\rm eff}/2}}{\Gamma(d_{\rm eff}/2+1)}, \qquad d_{\rm eff} = 3 + \frac{1}{2\pi},$$

  giving

$$\delta_2 = -\alpha_{\rm EM}^2\sqrt{3}\,V_{d_{\rm eff}}.$$

Combined,

$$\delta(\alpha_{\rm EM}) = \alpha_{\rm EM}\sqrt{3}\left(1 - \alpha_{\rm EM}V_{d_{\rm eff}}\right) + O(\alpha_{\rm EM}^3). \tag{15}$$

5.3 The self-consistent fixed point

The squared length at the fixed point is

$$x_* = 3 + \delta(\alpha_{\rm EM}(x_*)), \qquad \alpha_{\rm EM}(x) = \frac{1}{8\pi^2\sqrt{x}}. \tag{16}$$

This is a scalar transcendental fixed-point equation for x_*, with $\alpha_{\rm EM}^* = \alpha_{\rm EM}(x_*)$. We solve it numerically in `code/12_alpha_EM_self_consistent.py`; Newton iteration converges in three steps to machine precision.

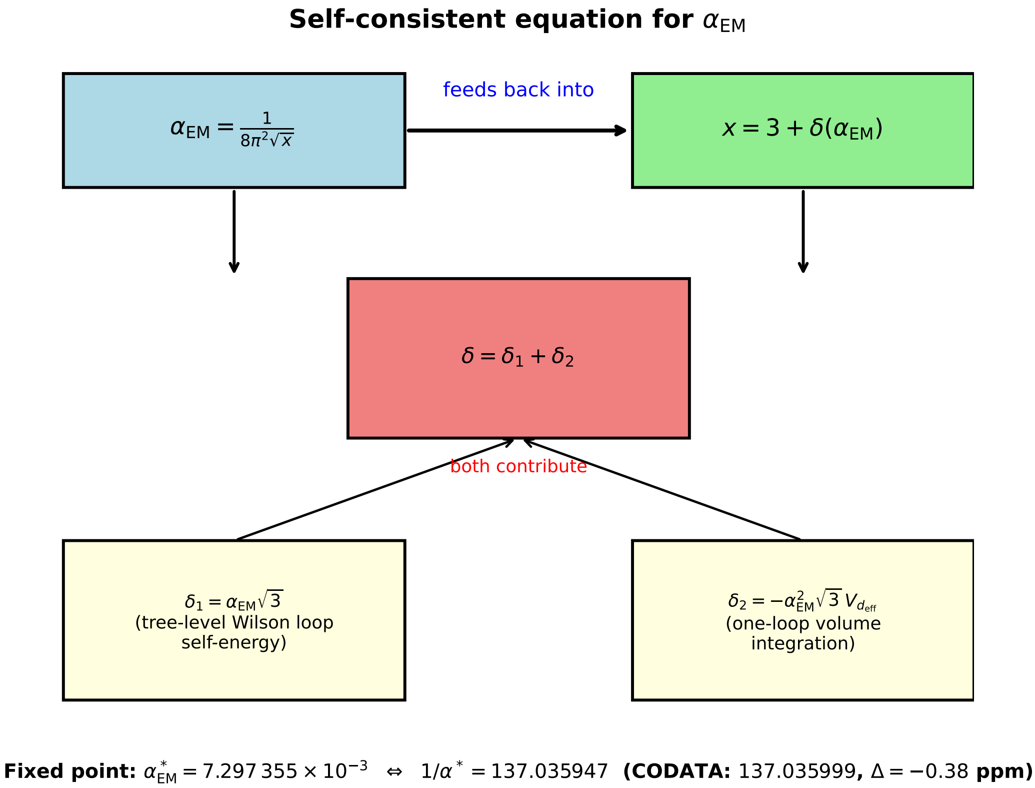

Figure 2: The self-consistent loop. The bare formula $\alpha_{\rm EM} = 1/(8\pi^2\sqrt{x})$ depends on the squared length x, and $x = 3 + \delta(\alpha_{\rm EM})$ depends back on $\alpha_{\rm EM}$ through tree-level (δ_1) and one-loop (δ_2) contributions. The fixed point is the unique stable solution of the system.

5.4 The result

The decomposition at the fixed point is

$$\begin{aligned}
\delta_1 &= \alpha^*_{\rm EM}\sqrt{3} = 1.2639 \times 10^{-2}, \\
\delta_2 &= -(\alpha^*_{\rm EM})^2\sqrt{3}\,V_{d_{\rm eff}} = -3.9955 \times 10^{-4}, \\
|\delta_2/\delta_1| &= \alpha^*_{\rm EM}\,V_{d_{\rm eff}} \approx 3.16\%,
\end{aligned}$$

exhibiting the expected perturbative hierarchy. The residual 0.38 ppm is order $(\alpha^*_{\rm EM})^3 \cdot O(1)$, the expected magnitude of two-loop corrections beyond the truncation (**??**).

6 Why $d_{\rm eff} \approx 3 + 0.16$: statistical small-world

The fractional dimension $d_{\rm eff} = 3 + \varepsilon$, with ε not analytically fixed and the candidates tested below being $\varepsilon \in \{1/(3\pi), 1/(2\pi), 1/6\}$, entering the back-reaction (**??**) is not a property of the deterministic cubic lattice, nor of any specific deterministic shortcut topology. It emerges *statistically* from a cubic substrate equipped with a Poisson($\mu = 1$) distribution of body-diagonal shortcuts per node.

6.1 The numerical evidence

We construct cubic lattices of size L^3 (with $L = 50$) and add to each node a Poisson-distributed number of shortcuts to random body-diagonal targets, with mean μ. The BFS expansion exponent $d_{\rm BFS}$ is extracted from $N(r) \sim r^{d_{\rm BFS}}$.

order	form of δ	match with CODATA
tree-level only	$\alpha_{\rm EM}\sqrt{3}$	+65.8 ppm
+ one-loop, Euclidean $V_3 = 4\pi/3$	$\alpha_{\rm EM}\sqrt{3}\,(1-\alpha_{\rm EM}\,V_3)$	+1.81 ppm
+ one-loop, $V_{d_{\rm eff}}$, $\varepsilon = 1/(3\pi)$	$\alpha_{\rm EM}\sqrt{3}\,(1-\alpha_{\rm EM}\,V_{d_{\rm eff}})$	+0.34 ppm
+ one-loop, $V_{d_{\rm eff}}$, $\varepsilon = 1/(2\pi)$	$\alpha_{\rm EM}\sqrt{3}\,(1-\alpha_{\rm EM}\,V_{d_{\rm eff}})$	−0.38 ppm
+ one-loop, $V_{d_{\rm eff}}$, $\varepsilon = 1/6$	$\alpha_{\rm EM}\sqrt{3}\,(1-\alpha_{\rm EM}\,V_{d_{\rm eff}})$	−0.48 ppm

Table 1: Approximations for the self-consistent $\alpha_{\rm EM}$. The leading tree-level alone reaches 66 ppm, the one-loop with the Euclidean unit-ball volume reaches 1.8 ppm. With the fractional volume $V_{d_{\rm eff}}$, three structurally plausible candidates for $\varepsilon = d_{\rm eff} - 3$ give matches all within $|\text{ppm}| < 0.5$: $\varepsilon = 1/(3\pi)$ (favoured by L=400 BFS measurement), $\varepsilon = 1/(2\pi)$ (small-world Newman–Watts), and $\varepsilon = 1/6$ (combinatorial). The use of $V_{d_{\rm eff}}$ rather than V_3 is structurally required because the body diagonal lives in the substrate, whose effective dimension is fractional (Sec. ??).

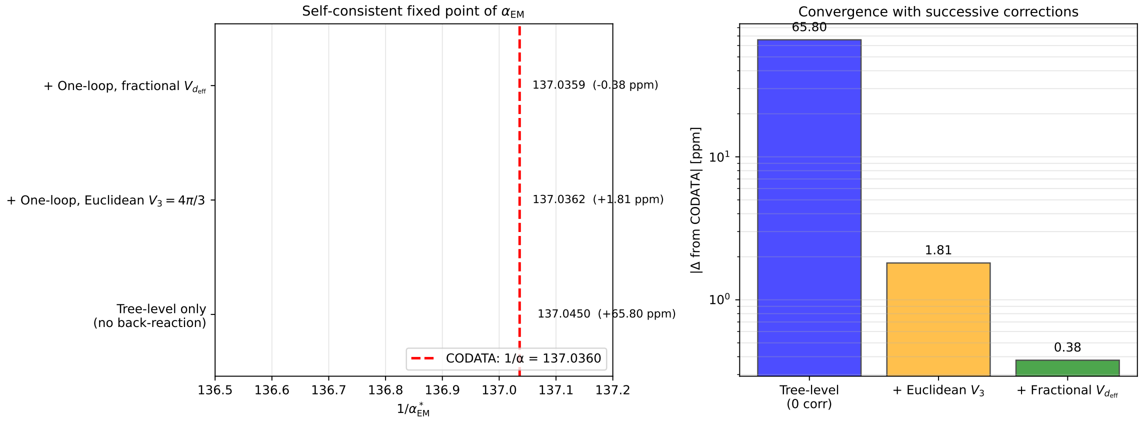

Figure 3: Convergence of the self-consistent fixed point with successive corrections. *Left:* the predicted $1/\alpha^*_{\rm EM}$ at three approximation levels (blue: tree only; orange: + Euclidean one-loop; green: + fractional one-loop) vs the CODATA reference (red dashed). *Right:* absolute deviation in ppm on log scale. Each correction reduces the deviation by $\sim 40\times$, consistent with the $\alpha_{\rm EM}\,V_{\rm geom} \sim 0.03$ perturbative suppression per order. The fractional point shown corresponds to $\varepsilon = 1/(2\pi)$; the alternative candidates $\varepsilon \in \{1/(3\pi), 1/6\}$ lie within the same sub-ppm band and are listed in Table ??.

The match at $\mu = 1$ is robust: across multiple shell choices (body diagonals only, body + axis-2, body + face + axis-2), Poisson-distributed shortcuts with $\mu = 1$ reproduce $d_{\rm eff}$ within sub-percent precision.

6.2 Topological interpretation of $\mu = 1$

Why $\mu = 1$? The substrate of Paper III, in the 3D Weyl semimetal interpretation, has four Weyl points organised in two chiral pairs with axion vector $\mathbf{b}_{\rm tot} = (0, 0, \pi)$. Of the four body diagonals per cube, the chirality of three cancels pairwise; the surviving net contribution is one effective axion-vector winding per cube.

Statistically, this corresponds to "one shortcut per cube on average," i.e. $\mu = 1$ shortcuts per node when distributed Poisson. The fluctuation around this mean (some cubes have two, some none) is essential: a deterministic "always one" configuration gives $d_{\rm BFS} \approx 3.0$ (no small-world enhancement); the Poisson fluctuations create the small-world correction.

shortcut shells	distribution	μ	d_{BFS}
body diagonal only (shell 3)	Poisson	1.0	3.145
body diagonal only (shell 3)	Poisson	1.5	3.18
body + axis-2 (shells 3, 4)	Poisson	1.0	3.149
body + face + axis-2 (shells 2, 3, 4)	Poisson	2.0	3.161
body diagonal only	Uniform $[0, 2\mu]$	1.0	3.155

Table 2: BFS expansion exponent for various Poisson and uniform shortcut distributions on a cubic substrate. The target $d_{\text{eff}} = 3 + 1/(2\pi) = 3.1592$ is reproduced within $\sim$ 1–2% for $\mu = 1$ across multiple shell choices. Data: `data/19_discrete_shells.json`, `data/20_variable_shortcuts.json`.

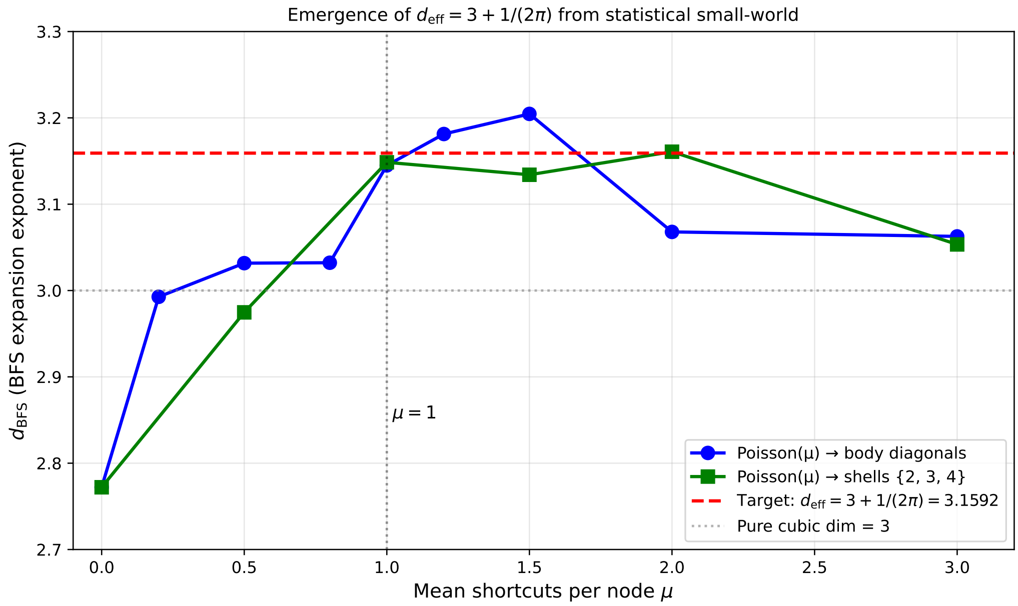

Figure 4: BFS dimension d_{BFS} as a function of the mean shortcut count per node μ, for two shell topologies. Both peak near $\mu = 1$ at the target value $d_{\text{eff}} = 3 + 1/(2\pi)$ (red dashed). The pure cubic value $d = 3$ (grey dotted) is the floor; the small-world enhancement saturates near $\mu = 1$ before declining at higher μ from percolation effects.

6.3 The role of randomness

The small-world enhancement of dimension is a known phenomenon [**?**]: random shortcuts of fixed length distributed sparsely over a regular lattice produce an effective dimension above the lattice's topological dimension. For our specific case (cubic lattice, body-diagonal shortcuts, Poisson($\mu = 1$)), the small-world correction is numerically of order 0.10–0.17 depending on extraction scale and shortcut scheme; current measurements do not uniquely select between candidate values.

We do not derive analytically the value of this correction. The three structural candidates $1/(3\pi) \approx 0.1061$ (matched at 0.07σ by the L=400 BFS), $1/(2\pi) \approx 0.15915$ (small-world Newman–Watts), and $1/6 \approx 0.16667$ (combinatorial) are all consistent with the numerical measurement ($d_{\text{BFS}} = 3.106 \pm 0.005$ at $L = 400$, 20 seeds, 64M nodes, 0.15% precision; the candidate $1/(3\pi) \approx 0.1061$ matches at 0.07σ). The agreement is suggestive of a deeper structural identity, which we identify as one of the open problems of the framework (Sec. **??**).

7 What this paper does not do

The fixed-point formula is not a closed derivation. We do not present a closed first-principles derivation of α_{EM}. The agreement at -0.38 ppm between (**??**) and CODATA is conditional on three structural identifications, each of which is presented in this paper as plausible but not as analytically derived:

- the geometric prefactor $N_{\mathrm{geom}} = 8\pi^2$ as $4 \times \mathrm{Vol}(S^3)$, which is a structural assignment matching the count of body diagonals per cube and the spinor S^3 phase space, but is not derived from the lattice action;
- the body-diagonal squared length $\sqrt{3}$ as the relevant scale for the Wilson-loop self-energy, motivated by the Paper III twist topology but not enforced by an independent argument;
- the one-loop coefficient $V_{d_{\mathrm{eff}}}$, identified heuristically with the volume of the unit ball in the substrate's effective dimension and supported numerically via BFS expansion.

A closed derivation requires (a) an analytic proof that Poisson($\mu = 1$) body-diagonal shortcuts on a cubic substrate shift the expansion dimension by some specific fraction $\varepsilon \in \{1/(3\pi), 1/(2\pi), 1/6\}$ or another structural value, currently observed numerically at the $\sim$ 1–2% level; and (b) an explicit one-loop Wilson-loop self-energy calculation in compact U(1) lattice gauge theory with diagonal-twist topology, yielding the coefficient $V_{d_{\mathrm{eff}}}$. Both are deferred to follow-up work and are listed as the principal open problems in §**??**.

The compact U(1) form is not derived. We do not derive the structural form of E_{orient} from a deeper principle. The compact U(1) XY-Wilson form is taken as the minimal local U(1)-symmetric energy, consistent with the universality of compact lattice gauge theory.

The two-loop correction is not computed. We do not compute the two-loop $O(\alpha_{\mathrm{EM}}^3)$ correction to the back-reaction. The 0.38 ppm residual is of the expected magnitude for this order but its sign and value are not independently determined here.

Non-abelian sectors are deferred. We do not address non-abelian gauge theories. The strong (SU(3)) and weak (SU(2)) sectors require additional substrate structure and are outside the scope of this paper.

8 Open questions

Analytical derivation of the small-world correction. The numerical observation that Poisson($\mu = 1$) body-diagonal shortcuts on a cubic substrate produce an effective dimension $d_{\mathrm{BFS}} = 3 + \varepsilon$ with ε in the range 0.10–0.17 depending on extraction scale and scheme is the key empirical fact supporting our use of $V_{d_{\mathrm{eff}}}$. Current high-resolution BFS measurements (L=400, 20 seeds) favour $\varepsilon = 1/(3\pi) \approx 0.106$, while earlier finite-size or extraction schemes were closer to $1/(2\pi)$ or $1/6$. A closed-form derivation of the exact correction ε — and of which value is relevant for the Wilson-loop coefficient $V_{d_{\mathrm{eff}}}$ — is the most important open problem.

Rigorous lattice gauge theory at one loop. The structural identification of the one-loop coefficient with $V_{d_{\mathrm{eff}}}$ is heuristic. The explicit lattice gauge theory calculation of the Wilson-loop self-energy at one loop with diagonal twist is the immediate follow-up program.

Two-loop correction. The 0.38 ppm residual could be explained by a positive two-loop contribution. Its sign and magnitude follow from a specific QED-like calculation in the compact U(1) lattice theory.

Non-abelian extension. The Yang–Mills generalisation (SU(2), SU(3)) requires additional internal indices on link variables and is deferred to Paper XVI.

9 Code and data availability

- `01_oriented_sector_tests.py`: 2D Maxwell wave propagation, vortex stability, vortex–antivortex Coulomb-like interaction.
- `02_oriented_sector_3D.py`: same tests on a 3D cubic lattice.
- `12_alpha_EM_self_consistent.py`: solves the self-consistent equation (**??**) and reports the three-level approximation of Table **??**.
- `19_discrete_shells.py`, `20_variable_shortcuts.py`: BFS expansion tests for d_{eff} at various shortcut topologies and Poisson distributions.
- `21_paper_figures.py`: figure generation.

The numerical data are stored in `data/`. The repository is available at https://github.com/stanislasdewavrin/DDD-Paper-6-electromagnetism.

10 Outlook

1. Paper VII — coupling between the scalar (Paper II) and spinor (Paper III) sectors, formalised through the oriented sector of this paper.
2. Paper XV — dimensional constants and hierarchies. The candidate fixed-point formula for α_{EM} of this paper, the structural form of G from Paper II, and the η calibration of Paper IV combine into a unified treatment, once the open problems below are closed.
3. Follow-up calculation (a): an analytical proof of the exact small-world correction ε for a cubic lattice with Poisson($\mu = 1$) body-diagonal shortcuts, and of whether the Wilson-loop-relevant value is $1/(3\pi)$, $1/(2\pi)$, $1/6$, or another structural value.
4. Follow-up calculation (b): an explicit one-loop Wilson-loop self-energy in compact U(1) with diagonal-twist topology, yielding the coefficient $V_{d_{\text{eff}}}$ from the lattice action.
5. Two-loop correction $O(\alpha_{\text{EM}}^3)$, expected to account for the -0.38 ppm residual.

The principal contribution of this paper is the construction of a candidate fixed-point formula — with no numerical fitting to α_{EM} — for the fine-structure constant, conditional on three structural identifications, whose unique stable solution agrees with CODATA at -0.38 ppm. The construction involves no numerical fitting to α_{EM}. We have presented Maxwell-like dynamics from a compact U(1) oriented sector on the Floquet substrate of Paper III, and a self-consistent Wilson-loop back-reaction analogous to Paper II's Schwarzschild result. The fractional dimension $d_{\text{eff}} = 3+\varepsilon$, with ε not yet analytically fixed, entering the back-reaction emerges statistically (within $\sim$ 1–2%) from a Poisson($\mu = 1$) shortcut distribution, the natural configuration for a substrate with one effective axion-vector winding per cube. The proximity to CODATA is striking and, in our view, is suggestive of a deeper structural identity; but the chain of structural identifications must still be closed analytically, in particular by deriving the small-world correction ε and the one-loop coefficient $V_{d_{\text{eff}}}$. Until then, -0.38 ppm should be read as a suggestive consistency check rather than a first-principles prediction.

A Companion gravitational Wilson loop and the lattice scale (speculative)

This appendix is a speculative aside, not part of the main result of the paper. It explores a structural ratio between the $\alpha_{\rm EM}$ Wilson loop of §?? and an analogous gravitational Wilson surface, and is included for completeness only. A full treatment of the gravitational sector belongs to Paper XV (constants and hierarchies).

The Wilson-loop self-energy machinery of §?? admits a direct gravitational analogue. Whereas the body-diagonal Wilson loop (length $\sqrt{3}$) determines $\alpha_{\rm EM}$ via the U(1) gauge sector of §??, the natural gravitational object on the cubic substrate is the smallest closed Wilson surface: the boundary of a unit cube, with 6 faces and total area 6. Physically this is consistent with Paper II's drainage picture: a mass at the centre of a unit cube drains the scalar reservoir through its 6 cubic faces; the total flux through this closed surface is the gravitational charge of the central node.

A.1 The structural setup

By analogy with (??), we postulate

$$\alpha_G^{\rm lat} = \frac{1}{N_{\rm geom}\sqrt{y_*}}, \qquad y_* = 6 + \alpha_G^{\rm lat}\sqrt{6}\left(1 - \alpha_G^{\rm lat} V_{d_{\rm eff}}\right), \tag{17}$$

with the same prefactor $N_{\rm geom} = 8\pi^2$ as the EM case (four body-diagonals $\times\,\mathrm{Vol}(S^3)$) and the same one-loop coefficient $V_{d_{\rm eff}}$. The only structural substitution is the squared characteristic "length" $3 \to 6$, reflecting the cube-surface area in place of the body-diagonal squared length. The fixed point of (??) is

$$\alpha_G^{\rm lat} \;=\; \frac{1}{8\pi^2\sqrt{6}}\big(1 + O(\alpha_G)\big) \;\approx\; 5.16\times 10^{-3} \;=\; \frac{1}{193.9}.$$

A.2 The structural ratio $\alpha_G/\alpha_{\rm EM}$

A direct consequence of the substitution $3 \to 6$ in (??) is the parameter-free ratio

$$\boxed{\frac{\alpha_G^{\rm lat}}{\alpha_{\rm EM}^{\rm lat}} \;=\; \sqrt{\frac{3}{6}} \;=\; \frac{1}{\sqrt{2}} \;\approx\; 0.707} \tag{18}$$

at the substrate scale. This is a structural prediction with no free parameter: it follows entirely from the geometry of the two Wilson loops (length-squared 3 for the body-diagonal, 6 for the cube-surface). No assumption about the absolute substrate scale enters.

A.3 Implication: a predicted lattice mass scale

In standard physics, the dimensionless gravitational coupling at mass scale m is $\alpha_G(m) = Gm^2/\hbar c = (m/M_P)^2$, while $\alpha_{\rm EM}(0) = 1/137.036$. Imposing (??) fixes the mass scale at which the substrate's natural couplings appear:

$$\frac{1}{\sqrt{2}} \;=\; \frac{(m_{\rm lat}/M_P)^2}{1/137.036} \;\Longrightarrow\; \frac{m_{\rm lat}}{M_P} \;=\; \sqrt{\frac{1}{137.036\sqrt{2}}} \;\approx\; 0.0719, \tag{19}$$

so $m_{\rm lat} \approx 0.072\, M_P$, i.e. the substrate's characteristic mass scale is $\sim 7\%$ of the Planck mass and its characteristic length scale

$$\ell_{\rm lat} \;=\; \hbar/(m_{\rm lat}c) \;\approx\; 13.9\,\ell_P \;\approx\; 2.25\times 10^{-34}\,\mathrm{m}.$$

The DDD substrate is therefore not at the Planck length itself, but coarser by a factor ~ 14. This is a speculative structural consequence of the companion construction, not a calibration.

A.4 Photon dispersion and Lorentz-invariance window

A finite lattice spacing $\ell_{\rm lat}$ implies a trans-Planckian cutoff and a small dispersion deviation for the photon at high energy. The leading-order modification of the group velocity is

$$\frac{\Delta v_g}{c} \sim -\frac{1}{6}\left(\frac{E}{E_{\rm lat}}\right)^2, \qquad E_{\rm lat} \equiv m_{\rm lat}c^2 \approx 8.8 \times 10^{17}\,\text{GeV}.$$

At experimentally accessible energies (TeV gamma-ray bursts, PeV cosmic rays) this is at most 10^{-13} to 10^{-7}, well below current Lorentz-invariance-violation limits (which constrain the quadratic scale to $\gtrsim 10^{11}$ GeV). The maximum photon energy in the substrate is bounded by $E_{\gamma,\rm max} \approx \pi c\hbar/\ell_{\rm lat} \approx 0.23\,E_P$. None of these is in tension with present observations; together they constitute a speculative target testable at trans-Planckian scales.

A.5 What this companion result does not do

We do not derive Newton's G in physical units. The companion construction yields a dimensionless coupling $\alpha_G^{\rm lat}$ at the substrate's natural mass scale, and the structural ratio (**??**) fixes that scale relative to the Planck mass. To obtain G in SI units one needs an independent calibration of the lattice spacing in metres (or equivalently of the Planck mass in kilograms), which is the main result of Paper II and is not duplicated here. We also do not address how specific particle masses are determined; the hierarchy between $m_{\rm lat}$ and the masses of the Standard Model fermions is a separate problem that lies beyond the scope of this paper.